Grammar–KT: Operationalising Grammar Knowledge Components for Knowledge Tracing

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Abstract

Knowledge tracing (KT) requires items, knowledge components (KCs), and an item–KC mapping, yet existing language-learning datasets rarely expose a transparent grammatical measurement structure. We present Grammar–KT, an auditable pipeline that separates canonical linguistic constructions (GRAMMARCELLS), a declared synthetic generator ontology (K^*, Q^*), observable items and responses, and downstream KC hypotheses $(\widehat{K}, \widehat{Q})$. All 1,222 source records receive an explicit disposition; 211 complete mappings yield 75 canonical cells. We freeze an outcome-independent 18-KC reusable-operation ontology, curate 113 model-screened items, construct a full-rank 113×18 Q^*, and simulate 283,000 replay-verifiable events from 1,000 learners. On this frozen benchmark, both coarsening and refinement worsen response prediction, while 10% Q-matrix corruption is reliably harmful. An observable-only KC-discovery procedure recovers a high-overlap representation but cannot distinguish atomic and compositional rules that have identical activation on seen items; unique recovery is therefore rejected even in a controlled positive case. Reusable KCs transfer substantially better than exact-cell KCs to 15 unseen combinations. The primary K^* ranking holds in every seed for 12 of 13 conditions (baseline plus 12 perturbations); unmodelled item difficulty is the exception. The release provides explicit construction evidence, public learner events, separate oracle truth, and controlled experiments—not a claim that the generator ontology or simulator describes human cognition.

1 Introduction

Knowledge tracing (KT) estimates changing learner knowledge from chronological responses (Corbett and Anderson, 1995; Abdelrahman et al., 2023). A KT dataset normally associates each item with one or more knowledge components (KCs), often through a binary Q-matrix. That matrix embeds consequential scientific decisions: what counts as reusable knowledge, which interactions require separate state, and where learning should transfer. It is not merely a data format.

Those decisions are unusually difficult to audit in language learning. The Teacher–Student Chatroom Corpus v2 (TSCCv2), for example, richly annotates 260 written English lessons, but a conversational turn may be explanation, scaffolding, elicitation, correction, or several grammatical phenomena rather than one predefined assessment opportunity (Caines et al., 2022). Vocabulary-oriented benchmarks do not isolate grammar KCs, while large tutoring logs may expose skill labels without a transparent linguistic decomposition. Scale alone therefore does not supply the measurement boundary needed for grammar-KC research.

Grammar resources offer richer descriptions, but not ready-made KT objects. The English Grammar Profile (EGP) contains over 1,200 corpus-derived competence descriptions (O'Keeffe and Mark, 2017). A descriptor can combine tense, aspect, voice, polarity, clause type, and modality, state bounded alternatives, or lie outside a declared scope. Moreover, one surface form admits competing KC analyses: *Why didn't he went?* could be one past-question KC, several reusable operations (finite-past selection, DO-support, negation, and non-subject-WH formation), or those operations plus an interaction. Each choice produces different learner histories and Q-matrices; no decomposition is neutral.

We introduce Grammar–KT, a controlled environment in which the scientific objects and their causal order are explicit:

$$\textsc{GrammarCell} \neq K^* \neq \widehat{K}. \qquad (1)$$

A GRAMMARCELL is a canonical linguistic description of the construction instantiated by an item. The generator inventory K^* and its deterministic item mapping Q^* are declared synthetic truth: simulated learners have mastery over these KCs. A discovered or deliberately misspecified inventory $\widehat{K}$ with mapping $\widehat{Q}$ is instead a downstream experimental hypothesis. None of these objects is asserted to be the cognitive truth of human grammar.

Figure 1 shows the resulting two-layer programme. Layer A processes the complete source inventory, freezes K^* before learner outcomes exist, constructs and audits a fixed measurement bank and Q^*, and only then simulates public responses plus separate private oracle truth. Layer B leaves that release immutable while changing or hiding the supplied KC representation.

We ask four questions:

RQ1 Can a grammar resource be transformed into an auditable fixed language-learning dataset with explicit KCs, items, item–KC mappings, and learner interactions?

RQ2 How does KT behave when the supplied KC representation differs from the generator ontology?

RQ3 Can observable learner-response evidence recover an appropriate KC representation when generator truth is hidden?

RQ4 How do alternative representations generalise from seen grammar to unseen combinations and unseen values?

Our contributions are: (i) a versioned benchmark constructed from every EGP record under an explicit English verbal-morphosyntax scope; (ii) an outcome-independent 18-KC generator ontology, 113-item bank, audited full-rank Q^*, and 283,000-event learner stream; (iii) controlled full-dataset studies of granularity, Q-matrix noise, discovery, linguistic generalisation, and mastery recovery; and (iv) compact simulator and collection-design controls that identify which assumptions carry the conclusions. The result is an executable methodological test bed. Automatic linguistic judgments and synthetic learner dynamics remain explicit limitations.

2 Related Work

Grammar resources and linguistic representation. The CEFR provides proficiency scales across language activities (Council of Europe, 2020). The EGP specializes this programme for English grammar, using learner-corpus frequency, accuracy, and dispersion to associate competence statements with CEFR levels (O'Keeffe and Mark, 2017). Recent work uses NLP to re-examine those level assignments quantitatively (Verratti-Souto et al., 2025). We do not reassess proficiency levels. Instead, we convert descriptor content into a bounded structural representation while retaining uncertainty, source relations, and out-of-scope decisions.

Language tutoring data and dialogue KT. TSCCv2 retains discourse sequences, pedagogical focus, response relations, and grammatical-error corrections from real English tutoring (Caines et al., 2022). Dialogue KT has used language models to mark candidate KCs and correctness at individual turns before fitting a tracer (Scarlatos et al., 2025). Such work demonstrates the value of expert-audited learner interaction, but it must infer whether a turn is an assessment opportunity and what it tests. We study the complementary setting: the linguistic opportunity, item, and generator mapping are constructed and audited before any learner event exists.

Exercise generation and documentation. Adaptive question generation has conditioned text on learner history and target difficulty (Srivastava and Goodman, 2021; Cui and Sachan, 2023). Grammar-focused systems generate multiple-choice or gap-filling exercises from passages or examples (Chan et al., 2022; Bitew et al., 2023). These studies highlight both authoring cost and the need to assess correctness and answer uniqueness. We generate common lexical contexts from fixed grammar cells, screen candidates independently, retain failed attempts and interventions, and freeze the bank before simulation. Following data-statement principles, we document scope, selection, and limitations (Bender and Friedman, 2018).

Domain models and knowledge tracing. KCs are acquired units connecting instruction and performance (Koedinger et al., 2012). Classical BKT tracks a latent learned/unlearned state with learning, guess, and slip parameters (Corbett and Anderson, 1995); factor models instead use logistic practice features (Pavlik et al., 2009). A Q-matrix supplies the assumed item–KC mapping, and learner responses can be used to validate or refine that mapping or compare cognitive models (de la Torre and

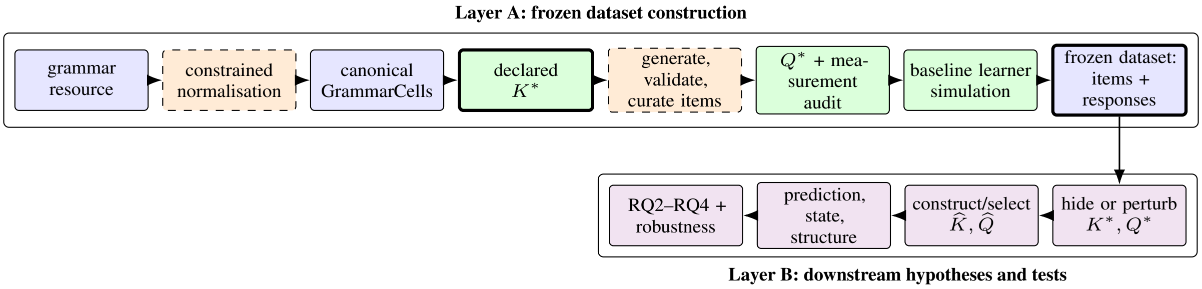

Figure 1: Grammar–KT separates dataset construction from downstream research. Generator truth is fixed before item responses exist. Later discovery and misspecification experiments may hide or perturb it but never redefine the immutable baseline. Model assistance is confined to the dashed linguistic normalisation and item-construction boxes.

Chiu, 2016; Pardos and Dadu, 2018; Cen et al., 2006). Related work combines domain-model induction with student modelling (Picones et al., 2022).

Our boundary differs in two ways. First, generator K^*/Q^* is fixed from linguistic and measurement declarations before responses are generated; it is not selected for predictive success. Second, response-driven refinement is evaluated later as KC discovery, where known synthetic truth can measure structural as well as predictive recovery. Synthetic KT data are useful for such controlled tests (Piech et al., 2015; Pagonis et al., 2024), provided simulator-dependent results are not promoted to human cognitive claims.

3 Scientific Objects and Source Data

3.1 Canonical grammar cells

The empirical scope is single-main-clause English verbal morphosyntax. For source descriptor d, constrained normalisation returns a status, one or more branches, explicit uncertainty, Phase-2 eligibility, and provenance. A scalar GrammarCell

$$c = (t, a, v, p, q, m) \tag{2}$$

contains tense, aspect, voice, polarity, clause type, and modal choice. Table 1 gives the domains. The value `NA` is a licensed structural value, not missing evidence. Compatibility rules include $m \neq$ none $\Rightarrow t =$ `NA` and imperative clauses requiring $t =$ `NA` and $m =$ none.

Phase 1 receives typed descriptor fields but no examples. It may return complete scalar branches, bounded alternatives, explicit missing constraints, or a zero-cell status. Phase 2 is called only for an eligible uncertain dimension with a licensed example; it may narrow named uncertainty but must preserve every Phase-1 branch and every ineligible exact value. Deterministic validation checks domains, statuses, compatibility, branch preservation, and resume-input identity. Only complete scalar mappings canonicalise. A separate relation table retains every descriptor-to-cell link, so repeated source evidence does not become a grammatical feature.

3.2 Complete source census

The consult-only input is a parsed 1,222-record EGP snapshot with a declared SHA-256 digest. We process every row rather than selecting descriptors likely to fit the schema. Descriptions outside the declared verbal-morphosyntax boundary remain auditable `out_of_scope` rows. This is full processing of the source under a bounded grammar scope, not coverage of all phenomena described by EGP.

The final census contains 211 complete, 327 partial, nine unresolved, and 675 out-of-scope mappings. Complete mappings yield 228 source–cell relations and 75 unique cells. A fresh balanced repeat of 120 Phase-1 rows gives 93.3% status agreement and 95.8% eligibility agreement. All 38 rows complete in both annotations have the same canonical cell set; partial-branch agreement is 79.0%. This supports explicit uncertainty rather than silent completion, but does not constitute expert annotation.

3.3 Semantic grammar regimes

Before simulation, a deterministic structural rule assigns 54 cells to `seen`, 15 to `unseen_combination`, and six to `unseen_value`. Each unseen combination contains only values and lower-order value pairs found in seen grammar, while its complete six-tuple is absent. The unseen-value cells contain the withheld perfect-progressive aspect value. All

Dimension	Symbol	Closed scalar domain	Principal constraint or interpretation
Tense	t	present, past, NA	NA is structural and differs from missing evidence
Aspect	a	none, progressive, perfect, perfect–progressive	Encodes auxiliary-chain aspect
Voice	v	active, passive	Passive is canonical BE-passive only
Polarity	p	positive, negative	Verbal polarity, not lexical meaning
Clause	q	declarative, polar question, subject-WH, non-subject-WH, imperative	Question subtypes and imperatives are distinct
Modal	m	none; can, could, may, might, must, shall, should, will, would	A central modal requires $t =$ NA

Table 1: The six dimensions of a scalar GRAMMARCELL. During partial normalization, a dimension may instead hold a finite list or null; only scalar cells enter canonicalization.

generator KCs nevertheless have seen support: this cohort tests a new linguistic composition, not a latent KC observed only at evaluation. Acquisition uses seen grammar only; every item receives one terminal, non-updating probe.

4 From Grammar to a Frozen KT Dataset

4.1 Outcome-independent generator KCs

Generator K^* is declared after canonicalisation and before item or learner outcomes exist. The selected reusable-operation hybrid contains 18 KCs: present and past finite-form selection; shared perfect and progressive dependencies; canonical BE-passive and verbal negation; imperative, polar-question, and non-subject-WH operations; and one KC for each of nine central modals. Active, positive, declarative, simple, modal-free, tenseless, and unsupported subject-WH conditions are not separate latent states. Perfect-progressive activates both shared aspect operations.

The declaration is stored as machine-readable activation rules. Generic projection code contains no English KC names. Before item construction, we also audited a 19-KC feature-only control, a hybrid with a perfect-progressive-chain KC, and a 75-KC exact-cell upper bound. The chain was rejected because its six-cell column is strictly nested in both components and adds no independently motivated operation; exact cells have no reuse. These decisions use linguistic coherence, reuse, support geometry, and parsimony, never response prediction.

4.2 Measurement-bank construction

For each of 75 cells, the default campaign draws three independent contextual controlled-production candidates from one model configuration. Generation receives the cell, English rulebook, item-format declaration, and a common-language instruction, but no learner outcomes, regime labels, KT metrics, or discovered KCs. A deterministic answer/slot check precedes an independent model validator. Acceptance requires target fidelity, grammaticality, naturalness, pedagogical suitability, determinacy, simple non-target language, no answer leakage, no extraneous grammar, and no unnecessary world knowledge.

Raw evidence is immutable. The default campaign left 18 cells uncovered, so we executed a frozen two-draw unchanged-prompt rescue, followed by a separately labelled explicit-construction intervention for the remaining determinacy-dominated cohort. An append-only accepted-answer correction fixed one finite omission. Open full-sentence imperative production remained indeterminate; a final preregistered all-and-only unordered-cue format bounded the response space while retaining imperative word order and negative DO-support. No repair is hidden as default generation.

Outcome-free curation retains at most two distinct accepted prompts per cell: the first and, where available, a contextually distant second candidate. This produces 113 unique prompts: 37 cells have one item and 38 have two. Max-one would retain 75 items; up-to-three adds 13 variants but no cell. The chosen bank supports contextual replication without treating surface variation as new grammatical structure.

Prompt and accepted-answer text are central to generation, validation, and the intended learner-facing measurement surface, but they have no causal role in the baseline response generator after the bank is frozen. The simulator neither renders a prompt nor collects or scores free text: an item identifier selects its Q^* row, mastery over the active KCs determines response probability, and keyed randomness samples binary y. Table 8 shows this boundary for one retained item. The baseline is consequently a controlled forward-generation and inverse-recovery benchmark, not a simulation of linguistic production.

4.3 True Q-matrix and structural gate

For fixed item i and generator KC k,

$$Q^*_{ik} = \mathbb{I}[k \text{ activates on } c_i]. \tag{3}$$

Projection is deterministic from the item cell and frozen KC rule; response data never enter. Before simulation, a mandatory gate measures item and cell support, row widths, KC co-occurrence, isolated opportunities, identical and near-identical columns, activation-equivalence classes, and matrix rank.

The final 113×18 Q^* has 269 edges, density .1323, rank 18, 75 distinct cell-activation rows, no identical columns, and no column pair with Jaccard similarity at least .90. Each KC has at least two items, although seven have fewer than six. Forty-six KC pairs have A-only, B-only, and A+B evidence; 105 have two-sided evidence without co-occurrence. The sole non-subject-WH cell leaves two nested pair geometries, retained as a source inventory limitation.

4.4 Baseline learner simulator

The simulator consumes only fixed items, K^*, Q^*, a protocol, and a seed. Learner-by-KC initial mastery is independently $\mathrm{Beta}(2, 2)$. For an item with active set A_i, the response probability is

$$p(y_i = 1) = .10 + .80 \min_{k \in A_i} m_k. \tag{4}$$

Every active KC receives outcome-independent opportunity learning $m'_k = m_k + .02(1 - m_k)$. The baseline has no forgetting, item difficulty, prerequisites, or transfer. Minimum aggregation was frozen because it is permutation invariant, noncompensatory, and invariant to repeating an equal-mastery prerequisite; product and compensatory means fail at least one of these interpretation checks.

A seen-only Q^*-balanced schedule gives each learner at least 12 opportunities per seen KC. It contains 170 acquisition events, followed by one non-updating probe of all 113 items. For 1,000 learners this yields 283,000 rows. Public events contain only learner, item, sequence, correctness, phase, pass, and grammar-regime fields. Active generator KCs, mastery before/after, response probability, and random draw are stored separately as private oracle truth.

4.5 Frozen release and downstream hypotheses

The versioned release records commands, configurations, hashes, deterministic gzip settings, and a recursive artifact inventory. An independent replay verifies every public/private link and state transition. Downstream studies may hide or perturb K^*/Q^* and construct $\widehat{K}$/$\widehat{Q}$, but cannot mutate the release. Pure representation comparisons reuse the same learner events. Oracle fields are opened only for explicitly labelled post-selection structural or mastery evaluation.

5 Downstream Experimental Design

Common observable protocol. All headline representation comparisons fit on the same 170,000 acquisition rows and evaluate the same 113,000 terminal probes. The primary model is a standardized PFA-like logistic regression (Pavlik et al., 2009) whose features use strictly prior successes and opportunities on active hypothesis KCs. It receives no item identity, generator state, response probability, random draw, or probe outcome. Log loss and Brier score are primary; calibration, AUC, and accuracy are diagnostic. Paired intervals resample whole learners 2,000 times, retaining all events for a sampled learner. Empirical history and fixed BKT are secondary sensitivities because their multi-KC semantics do not match every generator condition.

RQ2: controlled misspecification. Six deterministic representations span one all-merged KC, six linguistic family unions, the 18-KC K^*, 35 structural split-2 KCs, 66 split-4 KCs, and 75 exact-cell KCs. Separate controls add false-positive, false-negative, or mixed Q edges at a 27-edge (approximately 10%) Hamming budget over three structural seeds. Every projection is frozen before outcomes are loaded and all representations reuse the exact event stream.

RQ3: observable-only discovery. The outcome-independent candidate space contains 181 atomic features, operations, coarse unions, structural splits, supported intersections, exact cells, and deterministic hash distractors. Stable learner-disjoint groups provide 800 fit and 200 validation learners. A protected-feature forward selector scores validation log loss plus $.0005|K|$ and evaluates 22 eligible additions. Selection may read canonical item structure and seen acquisition outcomes, but not K^*, Q^*, oracle state, or any probe outcome. After freezing, name-free optimal column matching reports exact KC recovery, activation Jaccard, and aligned Q-edge precision, recall, and F1. The declared compositional operation policy is a candidate-space ceiling, not evidence of blind selection. RQ3's structural controls are local to the discovery protocol: its eight-KC operation-group baseline and 54-KC seen-cell baseline are not the six-family and 75-cell representations on the RQ2 granularity curve.

RQ4: linguistic generalisation. Six predeclared hypotheses—K^*, RQ3 atomic features, family unions, split-2, exact cells, and K^* plus supported intersections—are evaluated by seen, pairwise-seen/full-tuple-unseen combination, and unseen-value regimes. We report event-weighted and cell-macro metrics, per-cell and leave-one-cell-out sensitivity, and learner-paired contrasts. A separate negative control withholds one item from each of 30 two-item seen cells and replaces its acquisition occurrences with the same-cell counterpart while preserving every K^* opportunity count.

Mastery recovery. Observable predictions for the frozen RQ2 representations are written before the private oracle is opened. Under the known baseline response link, $(p - .10)/.80$ estimates the minimum active-prerequisite state and is compared with oracle pre-response aggregate mastery. A separate fixed-BKT analysis compares exposed terminal per-KC states with oracle KC mastery. The targets are intentionally distinct and are not called human mastery.

Simulator robustness. A compact preregistered study runs 13 conditions, three seeds, and 500 learners per world. It varies guess/slip, product and arithmetic-mean aggregation, learner noise and learning-rate heterogeneity, mild forgetting, item logit difficulty, correlated initial mastery, and correctness-conditioned updating. K^*, family-coarse, and split-2 projections share rows within each world. Primary conclusions use the unchanged observable logistic model; empirical and BKT results remain secondary.

Collection-design controls. Outcome-free nested cohorts test $N \in \{60, 120, 240, 500, 1000\}$; three common seeds test acquisition targets 6, 12, and 24. An outcome-free max-one/max-two audit separates within-cell replication from distinct Q rows. Finally, factorized A/B and planted A/B/interaction micro-worlds compare A+B-only, sparse-anchor, and balanced A-only/B-only/A+B banks at $N = 100, 300, 1000$ with 60 acquisition events and three terminal probes per learner. The latter distinguishes OR/union merging from an A+B-only intersection.

Model-assisted construction. Retained normalisation and generation use `gpt-5.6-sol`; independent item judgments use `gpt-5.6-terra`. Normalisation uses high reasoning effort and generation/validation medium. Rendered prompts, parsed outputs, settings, retries, and call evidence are retained. Provider sampling seeds and pinned snapshots were unavailable.

6 Results

6.1 RQ1: a frozen auditable dataset

All 1,222 source rows were processed without technical failure. Phase 2 is licensed for 106 partial rows, 105 of which have usable examples; it converts 41 to complete, leaves 57 partial, and makes seven unresolved. No remaining failure group justifies expanding the declared grammar schema. The exact cell-set agreement in the 120-row repeat is perfect among jointly complete cases, while lower partial-branch agreement supports retaining uncertainty.

Default $N = 3$ generation produces 225/225 valid payloads. The independent validator accepts 102 (45.3%), covering 57/75 cells. Determinacy fails for 117/223 model-judged candidates and is the dominant bottleneck; pedagogical suitability is next at 22 failures. Unchanged rescue adds nine cells and explicit-construction instructions add six. The finite answer-package correction succeeds for one of three candidates. Its two imperative failures are an informative negative result: open sentence production has an effectively unbounded valid response set. All four cue-bounded imperative candidates pass unchanged validation, yielding 75/75 coverage. These are automatic judgments, not expert gold.

Final source disposition	Count	Share
Complete	211	17.3%
Partial	327	26.8%
Unresolved	9	0.7%
Out of scope	675	55.2%
Source records	1,222	100%
Source–cell relations	228	–
Canonical cells	75	–

Item cohort	Candidates	Accepted	Cells added/covered
Default $N = 3$	225	102	57 covered
Unchanged rescue	36	10	9 added
Explicit construction	18	9	6 added
Answer-package copies	3	1	1 added
Cue-bounded imperatives	4	4	2 added
Accepted pool	–	126	75 covered
Fixed max-two bank	–	113	75 covered

Table 2: Full-source normalisation (left) and append-only item construction (right). Each intervention was frozen and separately labelled before its calls. Accepted denotes automatic independent validation, not human expert judgment.

Frozen artifact	Count
Canonical GRAMMARCELLS	75
Generator KCs	18
Fixed stored item records	113
Q^* dimensions	113×18
Q^* edges	269
Learners	1,000
Acquisition / probe events	170,000 / 113,000

Pre-simulation measurement diagnostic	Result
Q density	.1323
Column rank	18/18
Distinct cell activation rows	75
Identical / near-identical columns	0 / 0
Items per KC	2–49
KCs with fewer than six items	7
KC pairs with A-only, B-only, A+B	46
Two-sided pairs without co-occurrence	105

Table 3: The frozen full-v1 dataset and its outcome-free measurement gate. Near-identical means activation-column Jaccard similarity at least .90. Full rank does not eliminate rare-KC or finite-sample limitations.

The measurement gate passes before learner generation. Every KC has seen support, Q^* has full column rank, and there are no activation-equivalent or near-equivalent columns. The release contains 170,000 seen acquisition rows and 113,000 non-updating probes; an independent replay verifies all 283,000 public rows and their private state transitions.

6.2 RQ2: both coarsening and refinement hurt

K* obtains all-probe log loss .670627. On the frozen six-point curve, loss is minimized at K* and increases toward both coarser and finer representations: family merging costs .008132, structural split-2 .003165, split-4 .005868, and exact cells .015039. Every learner-paired interval in Table 4 excludes zero. No off-truth optimum was observed. This is a controlled predictive result, not a cognitive-truth result.

All nine 10% Q-corruption instances are harmful. Mean log-loss costs are .001685 for false positives (three-seed range .001265–.001970), .002644 for false negatives (.001791–.004004), and .002294 for mixed corruption (.001757–.002641). False negatives tend to cost more, but three structural seeds and overlapping ranges do not establish a universal ordering.

Oracle-only state evaluation agrees with the overall predictive ranking: K^* gives prerequisite-state RMSE .123738, versus .146300 family-coarse, .132752 split-2, .140428 split-4, and .163828 exact-cell. All four paired RMSE-difference intervals exclude zero. The conclusion is not uniform by regime: family-coarse improves RMSE by .003663 on the six unseen-value cells (95% interval $[-.005740, -.001507]$). A deliberately mismatched fixed BKT recovers unique learner–KC states poorly (RMSE .300804, correlation .434973), showing that a plausible KT state need not share the generator state's semantics.

6.3 RQ3: discovery reaches an equivalence class

The observable selector tests every eligible addition and retains its 18-feature base. Atomic features, compositional operations, and both automated projections have exactly the same seen-Q signature; separately fitted seen losses differ only numerically. After truth is opened, the compositional candidate-space ceiling exactly recovers all 18 generator activations. The selected atomic projection exactly recovers 16/18, with activation Jaccard .970854 and aligned edge F1 .965385. Its two aspect features fail to compose onto the six unseen perfect-progressive cells.

On all probes, atomic costs .000374 relative to the compositional ceiling (95% interval

Supplied representation	KCs	Probe log loss / Δ vs. K^* [95% CI]	State RMSE	Interpretation
All merged	1	$+.010225$ [.009380, .011029]	–	maximal coarsening
Linguistic families	6	$+.008132$ [.007450, .008779]	.146300	union merge
Generator K^*	18	.670627 (reference)	**.123738**	planted truth
Structural split-2	35	$+.003165$ [.002744, .003581]	.132752	context refinement
Structural split-4	66	$+.005868$ [.005281, .006473]	.140428	finer refinement
Exact cell	75	$+.015039$ [.014108, .015990]	.163828	no reuse

Table 4: RQ2 response prediction and oracle-only prerequisite-state recovery. Log-loss intervals resample 1,000 learners 2,000 times. State RMSE compares the inverse-linked public prediction with the oracle minimum active-KC mastery on 113,000 probes; all-merged was not in the frozen mastery subset.

Downstream hypothesis	KCs	Exact K^* KCs	Activation Jaccard	Aligned edge F1	Probe loss vs. ceiling
Compositional ceiling	18	18/18	1.000000	1.000000	.669606 (reference)
Selected atomic class	18	16/18	.970854	.965385	$+.000374$ [.000228, .000517]
RQ3 operation groups	8	5/18	.371913	.750929	+.005864
RQ3 seen-cell fine	54	1/18	.084184	.186969	+.006657
Hash distractors	18	0/18	.202342	.359259	+.013238

Table 5: RQ3 structural and predictive recovery after observable-only selection is frozen. Jaccard is padded optimal activation-column matching over the complete bank. The compositional row is a positive-control ceiling, not a blindly selected policy; only the selected-versus-ceiling delta is shown with its preregistered learner interval. The 8- and 54-KC controls are discovery-protocol baselines, distinct from the RQ2 6- and 75-KC conditions.

[.000228, .000517]), entirely through unseen-value extrapolation; seen predictions are identical. Hash distractors recover no exact KC and cost .013238. Adding supported intersections costs .000206 with interval $[-.000069, .000478]$, so that contrast is inconclusive. The candidate space contains the truth-like policy and rejects a spurious negative control, but observable seen responses cannot select uniquely among activation-equivalent rules. RQ3 therefore supports high-overlap equivalence-class recovery and rejects unique ontology recovery.

6.4 RQ4: reusable KCs transfer to new tuples

K* log loss is .669161 on seen grammar, .672036 on 15 pairwise-seen/full-tuple-unseen cells, and .681181 on six unseen-value cells. Exact-cell KCs cost .037609 on unseen combinations and .028627 on unseen values. Family unions and split-2 also have supported combination costs. Adding spurious intersections is harmful in all three regimes, empirically confirming that a union merge and an intersection are different perturbations.

Atomic and compositional predictions differ by at most 1.2×10^{-7} on seen and combination rows. Their unseen-value difference is inconclusive and changes point direction under the separate RQ3 fitting protocol; evaluation outcomes may expose the extrapolation but cannot legitimately break a tie formed on seen evidence. The unseen-value result is cell-sensitive: K* per-cell loss ranges from .666512 to .691208; leave-one-cell-out macro loss ranges from .680016 to .684955.

In the exact-item negative control, same-cell replacement preserves every K^* opportunity and all 30,000 held-out-item outcomes exactly match their paired baseline outcomes. The result follows from a simulator with no item memory or difficulty and outcome-independent same-Q updates; it does not imply human transfer between surface variants.

6.5 Supporting robustness evidence

The compact sensitivity study contains 39 worlds (13 conditions, three seeds) and 117 converged primary fits. In the baseline 500-learner condition, candidate-minus-K^* mean log-loss costs are .007924 for family-coarse and .003241 for split-2. K* beats both in every seed for 12/13 conditions, including altered noise, product or arithmetic-mean aggregation, learner noise/rate heterogeneity, mild forgetting, correlated initial mastery, and correct-only updates.

Unmodelled item logit difficulty (SD .60) is the exception. Split-2's mean cost remains .004138, but its three-seed range is $-.000413$–$+.009955$: it beats K^* once and falls behind the coarse representation once. K* ranks first in 38/39 primary worlds, so the headline is broadly robust but not invariant to item nuisance structure. Fixed BKT produces additional reversals under its known aggregation/update mismatch and does not drive the

Representation	Seen (54 cells)	Unseen combination (15)	Unseen value (6)
Generator K^* (absolute LL)	.669161	.672036	.681181
RQ3 atomic (delta)	≈ 0	≈ 0	$-.003236$ (n.s.)
Family union (delta)	$+.008940^\dagger$	$+.009185^\dagger$	$-.001751$ (n.s.)
Structural split-2 (delta)	$+.002234^\dagger$	$+.008806^\dagger$	$-.000684$ (n.s.)
Exact cell (delta)	$+.008209^\dagger$	$+.037609^\dagger$	$+.028627^\dagger$
K^* + intersections (delta)	$+.001415^\dagger$	$+.001196^\dagger$	$+.006954^\dagger$

Table 6: RQ4 event-weighted log loss by semantic grammar regime. Deltas are candidate minus K^* under one common fitting protocol. A dagger marks a 2,000-repeat whole-learner interval strictly above zero; n.s. marks an interval spanning zero. All unseen combinations are pairwise-seen and full-tuple-unseen.

conclusion.

In the collection controls, unpenalized prediction selects K^* in all 21 nested cohorts, but a fixed $.0005|K|$ criterion selects the family union in 18/21: the penalty changes the objective rather than consistently revealing generator truth. Raising the opportunity target from 6 to 24 improves K^* loss (.681757 to .636837) and widens its advantage. Max-two adds 38 within-cell variants over max-one and raises minimum KC support from one to two, but adds no Q row or rank.

With A+B-only micro-items, every representation ties exactly through $N = 1000$. Anchors restore full Q rank and make union merging detectably worse, yet omitting a planted interaction costs only .000506 under balanced anchors at $N = 1000$; the spurious-interaction negative control is approximately null. Thus response volume cannot repair structural equivalence, and full rank does not by itself guarantee practically unique predictive recovery.

7 Discussion

RQ1: construction is possible, but scope is part of the result. The pipeline turns a complete source census into the four essential KT artifacts without forcing every descriptor into one ontology. More than half the source is out of the declared single-clause verbal-morphosyntax scope, and uncertain mappings remain partial. That is preferable to a larger but untraceable inventory. The item campaign similarly exposes determinacy as a measurement problem and records the narrow intervention required for imperatives. Artifact integrity is strong; linguistic and pedagogical validity remain model-assisted rather than human-established. Because the synthetic y ignores prompt and answer strings, item quality affects the interpretation of each opportunity but cannot itself affect the generated response; the benchmark therefore cannot establish that a human could answer the stored exercise as intended.

RQ2: the planted representation is recoverable but conditional. In the baseline world, both union merges and structural refinements discard useful history, while exact cells cannot transfer. Q corruption has a smaller but consistent cost. Prediction and the item-level prerequisite-state target agree overall, yet the coarse unseen-value reversal shows why the best representation must name its estimand and regime. The item-difficulty robustness reversal further shows that a representation can absorb omitted nuisance variation and win for the wrong structural reason.

RQ3: predictive fit does not identify ontology uniquely. The discovery experiment is a controlled positive case: the candidate space contains an exact compositional ceiling, irrelevant hash candidates fail, and the selected atomic policy overlaps strongly with K^*. Nevertheless, atomic and compositional rules generate identical seen Q columns. No amount of additional response volume on those same activations can distinguish them. Predictive success is therefore insufficient evidence for cognitive KC truth; new measurement contrasts or linguistic priors are required to break the equivalence.

RQ4: reuse supports recombination, not unrestricted novelty. Reusable KCs transfer far better than exact-cell histories to 15 unseen six-tuples whose values and lower-order pairs are familiar. This is stronger than the earlier constituent-only notion of composition. The unseen-value cohort is narrower: all six cells concern perfect-progressive aspect, and its outcomes cannot be used to choose a policy that was tied on seen data. The result supports recombination under declared conditions, not general out-of-inventory language learning.

Implications for real data collection. The structural audit already distinguishes response volume from measurement diversity: repeated A+B items cannot identify A and B, whereas A-only, B-only, and A+B contrasts can. The robustness exception motivates crossed evidence for item difficulty rather than allowing KC columns to absorb item nuisance. Any learner-count or opportunity threshold from the collection-design controls must be reported as a conditional synthetic failure boundary, not a human sample-size prescription. The controls sharpen the distinction: a second within-cell variant adds replication but no activation contrast, anchors make a union merge visible, and even a full-rank bank leaves a weak planted interaction difficult to recover. Rank must therefore be paired with effect size and uncertainty.

Language-independence boundary. Projection, Q auditing, simulation, merge/split transformations, KT fitting, and evaluation are schema-driven. A toy schema using `mood` and `person` executes cells through K*, Q, simulation, and public events without English branches. EGP parsing, the six dimensions, English KC declarations, and item prompts remain language-specific. This establishes a software abstraction, not cross-lingual empirical validity.

8 Reproducibility and Audit Trail

The frozen release is rooted at `data/grammar_kt_full_v1/`. Its top-level manifest distinguishes linguistic representation, generator truth, observable data, and private oracle artifacts. It records the pre-response code revision, exact simulator command, configuration and input hashes, deterministic gzip metadata, row counts, and an 88-file recursive inventory. The observable and private-oracle SHA-256 values begin `9272ca86a647` and `956ed53f370d`, respectively; the complete values are in the manifest and evidence ledger.

`scripts/build_dataset.py` is the chronological Layer-A runner. It loads the typed source, normalises and canonicalises, constructs generator KCs, generates/validates/curates items, builds and audits Q^*, and invokes the frozen simulator. Q and dataset verification modes replay deterministic artifacts without making LLM calls. Raw prompts and outputs, independent judgments, intervention plans, and immutable superseded records are linked through provenance directories.

Separate scripts under `scripts/experiments/` implement misspecification, discovery, generalisation, mastery recovery, robustness, and collection design. Every substantive study writes its question, hypothesis, manipulation, held-fixed variables, seeds, input hashes, and intended interpretation before its final run. Pure representation comparisons verify a shared probe-event hash. Discovery freezes selection before truth is loaded; mastery recovery freezes observable predictions before opening the oracle. Exact commands and artifact hashes are retained in the experiment ledger.

The public stream is sufficient for ordinary KT and KC discovery. Oracle state is not embedded in public rows or required by the fitting code. A non-English-named toy schema exercises the generic cells-to-K*-to-Q-to-events contract. Unit and scientific-contract tests cover leakage boundaries, non-updating probes, deterministic Q projection, alternate-schema execution, same-event representation comparisons, replay hashes, and all paper-facing experiments.

9 Limitations and Ethics

Linguistic scope and annotation. The empirical domain is English single-main-clause verbal morphosyntax under six declared dimensions. All EGP records are processed, but 675 are outside that boundary and only complete scalar mappings canonicalise. Normalisation is model-assisted; a 120-row repeat measures stability but there is no expert adjudication or second-language study. The external parsed source snapshot is consult-only and not redistributed.

Item validity. Items are LLM-generated and independently model-validated. No teachers or learners assessed correctness, naturalness, difficulty, accessibility, bias, or educational value. Determinacy caused most rejection, and two imperative cells required a separately labelled cue-bounded format after open production and a finite answer-list correction failed. The fixed bank is auditable measurement material, not a classroom-ready assessment. Prompt and accepted-answer strings are never rendered to the simulated learner or used to score y, so ambiguity cannot affect the synthetic outcome—but neither can this benchmark test whether a person would interpret or answer an item as intended.

Generator truth. The 18 KCs are a deliberately simple reusable-operation ontology. They are ground truth only because the simulator maintains state over them. Reference conditions, excluded chain operations, minimum aggregation, independent initial mastery, and equal opportunity learning are scientific assumptions, not evidence about cognitive atomicity or human acquisition.

Synthetic learners and uncertainty. The baseline uses $\mathrm{Beta}(2,2)$ mastery, guess/slip .10/.10, learning rate .02, and no item difficulty, forgetting, vocabulary state, prerequisites, or transfer. Compact sensitivity worlds vary these assumptions mostly one at a time, apart from the combined guess/slip design, and use three seeds and one severity per condition. The correlated-mastery control is undirected rather than a prerequisite graph. Learner bootstrap intervals quantify sampling of synthetic learners within a world, not uncertainty over grammar cells, simulator families, prompts, or human populations.

Evaluation scope. The six unseen-value cells all test perfect-progressive aspect, and every generator KC has seen support. They do not establish generalisation to arbitrary unseen grammar. The exact-item negative control is implied by the absence of item memory and difficulty. Family-coarse state recovery reverses locally on the unseen-value cohort, and unmodelled item difficulty reverses one primary split-2 ranking, so K^* is not claimed to dominate every regime or world.

KC discovery and KT models. Activation equivalence is bank-specific. Seen responses cannot distinguish atomic and compositional rules with identical seen Q columns; a structural prior or new contrast is needed. The compositional ceiling proves candidate reachability, not blind recovery. PFA-like logistic regression is the primary observable model; fixed BKT uses mismatched aggregation and update semantics and is diagnostic only. Similar response prediction does not guarantee recovery of a meaningfully comparable latent state.

Collection and language claims. Synthetic learner-count, opportunity, or anchor thresholds do not transfer directly to human-study recruitment. They identify failure modes under stated generative conditions. The alternate-schema test verifies reusable software, not empirical cross-lingual validity or a universal grammar ontology.

Ethics. No human-participant or learner records are used; all response sequences are synthetic. Latent conditions denote simulation assumptions, not demographic or ability groups. Generated items must not be deployed for assessment or instruction without qualified human review for correctness, ambiguity, accessibility, and bias. Use and quotation of EGP material remain subject to its licence; the release retains derived mappings and verification metadata rather than redistributing the consult-only source snapshot.

10 Conclusion

Grammar–KT demonstrates that a grammar resource can be transformed into an auditable KT benchmark when canonical constructions, generator knowledge, measurement, and downstream hypotheses remain distinct. The full-v1 release contains explicit K^*, fixed items, deterministic Q^*, observable responses, separate oracle truth, and construction provenance.

Known truth then supports questions opaque datasets cannot answer. Both coarsening and refinement harm prediction in the baseline; exact-cell KCs transfer poorly to unseen grammatical tuples; and observable KC discovery recovers a strong equivalence class without uniquely identifying the planted ontology. State recovery and simulator sensitivity show why prediction alone is not enough: KT state semantics can mismatch generator mastery, and item difficulty can reverse a representation ranking. These are controlled methodological findings and design guidance for future learner collection, not claims about the cognitive decomposition of English or the behaviour of human learners.

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A Artifact Map

Scientific object	Inspectable declaration and frozen evidence
Source and GRAMMAR-CELLS	modules/grammar/resource/egp/; modules/grammar/canonical/; data/grammar_kt_full_v1/grammar/; normalisation and inventory provenance
Generator K^*	modules/kcs/generator/; data/grammar_kt_full_v1/kcs.jsonl; construction provenance
Item bank	modules/items/; data/grammar_kt_full_v1/items/items.jsonl; immutable generation, validation, correction, and curation evidence
True Q^*	scripts/build_true_q_matrix.py; data/grammar_kt_full_v1/q_matrix.csv; data/grammar_kt_full_v1/oracle/q_matrix_sparse.jsonl; data/grammar_kt_full_v1/provenance/measurement/audit.json
Observable and oracle events	modules/simulation/; scripts/freeze_baseline_dataset.py; data/grammar_kt_full_v1/interactions.jsonl.gz; data/grammar_kt_full_v1/oracle/learner_truth.jsonl.gz
Downstream hypotheses	scripts/experiments/; experiments/full_v1/; reports/full_v1_artifacts/
Release and claims	data/grammar_kt_full_v1/manifest.json; reports/final_release_manifest.json; reports/experiment_log.md; ACL/evidence.md

Table 7: Repository-root map from scientific objects to their inspectable declarations and frozen evidence.

B Generator and Measurement Detail

The 18 generator KCs fall into five auditable groups:

- finite-form selection: present and past;
- aspectual dependencies: perfect and progressive;
- voice/polarity: BE-passive and verbal negation;
- clause operations: imperative, polar question, and non-subject-WH; and
- central-modal identity: can, could, may, might, must, shall, should, will, and would.

Seventeen recur across cells. Seven have fewer than six final items, but every KC has at least two items and 12 seen acquisition opportunities per learner. The perfect-progressive-chain alternative would be active in six cells but is nested in both aspect dependencies; it was excluded before item generation.

The final bank contains 84 seen, 20 unseen-combination, and nine unseen-value items. Max-one preserves all 75 cells with 75 items, while max-three adds only 13 items beyond max-two. No selected prompts are exact duplicates. Surface nonidentity is not itself evidence of pedagogical diversity.

C Research-Question Ledger

RQ1—supported in scope. All 1,222 rows receive a disposition; the complete subset yields 75 cells, 18 generator KCs, 113 items, audited Q^*, and 283,000 replayable events.

RQ2—supported for frozen controls. K^* gives the lowest overall primary loss and item-state RMSE; coarsening, refinement, and every tested 10% Q corruption are harmful. Local and robustness reversals delimit the claim.

RQ3—unique recovery rejected. Observable selection recovers a high-overlap seen-Q equivalence class, but cannot distinguish atomic from compositional rules with identical seen activation.

RQ4—recombination supported. Reusable KCs transfer better than exact cells to pairwise-seen/full-tuple-unseen grammar. The six-cell unseen-value comparison is narrow and cannot resolve RQ3 equivalence.

Every answer is conditional on the declared synthetic world and automatically screened item bank.

Field	Frozen value and role
Item ID	`candidate_gc_e7fef77abc10b5ba_01`
Stored prompt	*Mia had a map, so she found the house. Without the map, [____]. (find)*
Stored answer key	*she would not have found the house*; or *she wouldn't have found the house*
GRAMMARCELL	`tense=NA, aspect=perfect, voice=active, polarity=negative, clause=declarative, modal=would`
Active K^* / Q^* row	perfect construction; central modal WOULD; verbal negation / `100000000000001100` (K01–K18)
Example public event	simulated `learner_000001`, sequence 2, acquisition, y=1
Causal role in baseline	The item ID selects the Q^* row. Active-KC mastery and keyed randomness generate binary y; no prompt is rendered and no textual response is collected or scored.

Table 8: One retained item and event, separating the auditable measurement surface from response generation. The stored answer is not evidence that the simulated learner produced it verbatim.

Generator condition	Family-coarse Δ LL		Split-2 Δ LL	
	Mean	Three-seed range	Mean	Three-seed range
Baseline minimum, .10/.10	.007924	[.007568,.008306]	.003241	[.003001,.003575]
Noise .00/.00	.016363	[.016032,.016957]	.005969	[.005573,.006636]
Noise .20/.10	.005259	[.004938,.005684]	.002222	[.001971,.002457]
Noise .10/.20	.005310	[.005110,.005692]	.002283	[.002159,.002526]
Noise .20/.20	.003082	[.002904,.003436]	.001506	[.001372,.001604]
Product aggregation	.008237	[.007820,.008762]	.004188	[.003755,.004678]
Arithmetic-mean aggregation	.004500	[.004353,.004790]	.001738	[.001486,.002218]
Learner guess/slip heterogeneity	.008186	[.007748,.008794]	.003294	[.003077,.003722]
Forgetting .002/gap	.003467	[.003061,.003829]	.001753	[.001567,.001979]
Item difficulty logit SD .60	.006346	[.004579,.007505]	.004138	[-.000413,.009955]
Learner learning-rate heterogeneity	.007939	[.007471,.008279]	.003311	[.002908,.003588]
Correlated initial mastery	.005836	[.005357,.006275]	.002380	[.002034,.002737]
Correct-only updating	.011085	[.010698,.011391]	.005046	[.004769,.005209]

Table 9: Simulator-sensitivity detail for the primary observable-logistic model. Values are candidate minus K^* on common terminal probes; positive favours K^*. Seeds are 20260829–20260831 with 500 learners per world. Ranges describe these three structural seeds, not population intervals.

Control	Frozen result	Interpretation boundary
Nested learners	Raw prediction selects K^* in 21/21 cohorts; $.0005\|K\|$ selects union in 18/21.	Selection depends on the criterion; no synthetic count is a human recruitment threshold.
Opportunities 6/12/24	K^* loss is .681757/.672104/.636837; its gap to exact cells grows .007523/.014301/.032110.	Schedule lengths differ; no diminishing-return threshold is inferred.
Max-one vs. max-two	75 vs. 113 items and minimum support 1 vs. 2, but both have 75 Q rows and rank 18.	Added contexts are replication, not structural diversity in this simulator.
Two-KC anchors	A+B-only gives exact ties through $N = 1000$; anchors restore rank, but missing planted interaction costs only .000506 at balanced $N = 1000$.	Rank is necessary but insufficient for practically unique predictive recovery.

Table 10: Bounded collection-design controls. All counts and effects are conditional on the declared synthetic worlds.